

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary and Advanced Level

MARK SCHEME for the November 2004 question paper

9702 PHYSICS

9702/06

Paper 6, maximum mark 40

This mark scheme is published as an aid to teachers and students, to indicate the requirements of the examination. This shows the basis on which Examiners were initially instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began. Any substantial changes to the mark scheme that arose from these discussions will be recorded in the published *Report on the Examination*.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the *Report on the Examination*.

- CIE will not enter into discussion or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the November 2004 question papers for most IGCSE and GCE Advanced Level syllabuses.



Grade thresholds taken for Syllabus 9702 (Physics) in the November 2004 examination.

	maximum mark available	minimum mark required for grade:		
		A	B	E
Component 6	40	30	27	15

The thresholds (minimum marks) for Grades C and D are normally set by dividing the mark range between the B and the E thresholds into three. For example, if the difference between the B and the E threshold is 24 marks, the C threshold is set 8 marks below the B threshold and the D threshold is set another 8 marks down. If dividing the interval by three results in a fraction of a mark, then the threshold is normally rounded down.



November 2004

GCE A AND AS LEVEL

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 9702/06

PHYSICS
Paper 6



UNIVERSITY of CAMBRIDGE
International Examinations

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Option A – Astrophysics and Cosmology

- 1 diameter of the Sun B1
nearest (neighbour) star/Proxima Centauri B1
diameter of (Milky Way) galaxy B1
extent of (visible) Universe (allow diameter/radius) B1 [4]
- 2 e.g. Atmospheric absorption/scattering M1
means light is too faint A1
Light pollution M1
means light cannot be distinguished against background A1 [4]
Irregular atmospheric refraction/thermal currents (M1)
means small objects blurred/not seen (A1)
(any two sensible suggestions {M1 x 2} plus some further detail of each {A1 x 2})
- 3 (a)(i) either density such that Universe will not collapse or expand indefinitely B1
greater density than ρ_0 means collapse (OR vice versa) B1
or determines whether Universe is 'open' or 'closed' (B1)
greater density than ρ_0 means 'closed'
OR smaller density than ρ_0 means 'open' (B1) [2]
- (ii) (if Universe is closed eventually all) kinetic energy of galaxies will be converted to B1
(gravitational) potential energy B1 [2]
(gravitational) potential energy involves the gravitational constant G
- (b)(i)1 (sensible straight line and) one or two points chosen with attempt at antilogs B1
 $H_0 = 100 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (allow $80 \rightarrow 125 \text{ km s}^{-1} \text{ Mpc}^{-1}$) A1
 $1 \text{ Mpc} = 3.1 \times 10^{19} \text{ km}$ C1
 $H_0 = 100 / (3.1 \times 10^{19}) = 3.2 \times 10^{-18} \text{ s}^{-1}$
Age = $1/H_0 = 3.1 \times 10^{17} \text{ s}$ A1 [4]
- (i)2 $\rho_0 = (3 \times 10^{-18})^2 / (8 \times \pi \times 6.67 \times 10^{-11})$ C1
 $= 1.86 \times 10^{-26} \text{ kg m}^{-3}$ A1 [2]
- (ii) number density = $(1.86 \times 10^{-26}) / (1.66 \times 10^{27})$ C1
 ≈ 10 A1 [2]

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Option F – The Physics of Fluids

- 4 (a) M shown near base of stem B1 [1]
- (b)(i) density = mass/volume C1
volume submerged in liquid of density $1.0 \text{ g cm}^{-3} = 165 \text{ cm}^3$ C1
volume submerged in liquid of density $1.1 \text{ g cm}^{-3} = 150 \text{ cm}^3$ C1
change in volume = 15 cm^3 A1
- (ii) distance (= $15/0.75$) = 20 cm A1 [5]
- 5 (a) arrows longer at centre than edges M1
arrows parallel and correct relative lengths A1 [2]
- (b)(i) no unique value of (linear) speed B1 [1]
(ii)1 volume flow rate doubles A1
(ii)2 new radius = $1.05 r$
new flow rate = 1.054×2 C1
= 2.4(3) times greater A1 [3]
- 6 (a) (fluid) flow/movement B1
that is erratic/has eddies B1
i.e. speed varies continuously (in magnitude and direction) with time B1 [3]
- (b)(i) for turbulent flow, F_D/v^2 C1
 $v = 58 \text{ m s}^{-1}$ A1 [2]

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Option M – Medical Physics

- 7 (a)** pulse of ultrasound
reflected from boundaries
received (at surface) and processed
time for pulse to return gives depth of boundary
reflected intensity gives information on nature of boundary
B1
B1
B1
B1
B1 [5]
- (b)** fraction = $e^{-23 \times 0.055}$
= 0.28
C1
A1 [2]
- (c)** fraction = $0.28 \times 0.35 \times 0.28$
= 0.027
(or $0.35e^{-23 \times 0.11} = 0.028$)
C1
A1 [2]
- 8 (a)(i)** rays from S converge to point behind retina
B1
- (ii)** range of image distances
such that image is tolerably in focus
B1
B1 [3]
- (b)** for the same size of patch on the retina
focused image is further from the retina
(so) depth of focus is increased
M1
A1
B1 [3]
- 9 (a)** intensity = $(0.33 \times 10^{-6}) / (65 \times 10^{-6})$
= $5.1 \text{ (5.08)} \times 10^{-3} \text{ W m}^{-2}$
I.L. = $10 \lg (5.08 \times 10^{-3}) / (1.0 \times 10^{-12})$
= 97 dB
C1
C1
C1
A1 [4]
- (b)** (long-term exposure) could cause deafness OR
(short-term exposure) could cause tinnitus
B1 [1]

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Option P – Environmental Physics

- 10 (a)** massive nucleus/named appropriate nucleus splits into two approximately equal parts/named components with the release of neutrons and energy
B1
B1
B1 [3]
- (b)** moderator: slows down (high speed) neutrons so that further fissions are more likely/will take place
control rods absorb neutrons to provide control over the rate of fission
M1
A1
M1
A1 [4]
- 11 (a)(i)** water moved from (area of) trough to crest to form wave
potential energy = mgh
 $= \frac{1}{2} \lambda A w \rho \times g \times A$
(must be laid out so that substitutions are obvious)
 $= \frac{1}{2} w A^2 \lambda \rho g$
B1
M1
M1
A0 [3]
- (ii)** there are V/λ wavecrests passing a point per unit time
power = $\frac{1}{2} w A^2 \lambda \rho g \times V/\lambda$
 $= \frac{1}{2} w A^2 \rho g V$
M1
A1
A0 [2]
- (b)** e.g hazard to shipping, unsightly, upset to shoaling fish etc. (any sensible suggestion)
B1 [1]
- 12 (a)** input shown clearly as 1140 W
four outputs labeled correctly
arrows having approximately correct ratio of widths
B1
M1
A1 [3]
- (b)** electrical heating more efficient at transferring energy to water
very little thermal energy escapes because plastic is an insulator
gas ring much less efficient because of thermal energy losses to the air
thermal energy losses due to conduction as kettle is metal
B1
B1
B1
B1 [4]

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Option T – Telecommunications

- 13 (a)** box for 1 m – 10 cm labeled T B1
- (b)** box for 10 cm – 1 cm labeled S B1 [2]
- 14 (a)** frequency of carrier wave varies (in synchrony) with information signal B1
constant amplitude OR carrier frequency >> signal frequency B1
change in frequency measures displacement of information signal B1
rate at which carrier frequency varies gives frequency of information signal B1 [4]
- (b)(i)** period = 0.8 μ s C1
frequency = 1.25 MHz A1
- (ii)** 125 kHz A1 [3]
- (c)** advantage: e.g. better quality/less interference B1
disadvantage: e.g. more transmitters/more expensive B1 [2]
(any sensible suggestions, 1 each)
- 15 (a)(i)** sampled every 0.5 ms C1
frequency = 2.0 kHz A1
- (ii)** at 1.0 V intervals B1
- (iii)** 4 bits B1 [4]
- (b)** needs sampling time shorter than smallest peak-trough interval B1
any suggestion of about (0.2 ms or about) 5 kHz (*allow 5 kHz $\rightarrow$ 10 kHz*) A1
needs voltage interval less than peak-trough height B1
any suggestion at about 0.3 V (*allow 0.1 V $\rightarrow$ 0.4 V*) C1
so either $12/0.3 = 40$ OR $11/0.3 = 37$ OR $10/0.3 = 34$ etc. A1
(ignore binary nature of the ADC and the DAC)